Chapter 8, Lecture 4

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Method of Maximum Likelihood

Example. Suppose that we are given $X = 5$ from a Poisson(μ) distribution with $\mu > 0$ unknown.

Q: What is the *most likely* value for μ?
A: The probability that we observed $X = 5$ is given by the formula

$$\mathbb{P}\{X = 5\} = \mu^5 \exp(-\mu)/120$$

For instance, We see that $\mathbb{P}\{X = 5\}$, as a funtion in μ, is maximized at $\mu = 5$. That is,

μ	1	2	3	4	5	6	7
$P(X = 5)$	0.0031	0.0361	0.1008	0.1563	0.1755	0.1606	0.1277

the most likely parameter to have generated the observation $X = 5$ is $\mu = 5$. Of course, more rigorously, we observe that maximizing $\mathbb{P}\{X = 5\}$ is the same as maximizing $\log(\mathbb{P}\{X = 5\})$ because the log is a strictly monotone increasing function. We find

$$\log \mathbb{P}\{X = 5\} = 5 \log(\mu) - \mu - \log(120)$$

and taking a derivative with respect to μ, and setting this equation to zero yields

$$\frac{5}{\mu} - 1 = 0 \Longrightarrow \mu = 5.$$

There is of course a general principle behind this.

General maximum likelihood principle

Let $\mathbf{X} = (X_1, \ldots, X_n)$ have a joint density function (pdf) or probability mass function (pmf) $f(\mathbf{x}|\theta)$. The *likelihood function*

$$\text{lik}(\theta) := f(\mathbf{x}|\theta) = f(x_1, \ldots, x_n|\theta)$$

is simply the joint pdf or pmf as a function in the parameter θ, given the data.
In the discrete case, the likelihood gives the probability of observing the given data as a function in θ.
A similar argument can be made for the continuous case as

$$P(\mathbf{X} \in (\mathbf{x}, \mathbf{x} + \delta)) \approx \delta f(\mathbf{x}|\theta) = \delta \text{lik}(\theta)$$

The *maximum likelihood estimator* (MLE) is the value of θ that maximizes the likelihood, that is, it gives the most likely value of the parameter that resulted in the observed data. That is,

$$\widehat{\theta}_{MLE} = \arg\max_{\theta} \text{lik}(\theta)$$

For now, there is no guarantee that the MLE is unique, or that it even exists.

As already seen in the example, maximizing the likelihood is the same as maximizing the log-likelihood. The latter is oftentimes easier to compute. Lastly, in the special case that $\mathbf{X}$ consists of n iid observations $X_1, \ldots, X_n$ from $f(x|\theta)$, we have that

$$\text{lik}(\theta) = \prod_{i=1}^{n} f(X_i|\theta).$$

We now give a few examples.

Poisson(μ)

We have

$$\text{lik}(\mu) = \mu^{\sum_{i=1}^n X_i} \exp(-n\mu) / \prod_{i=1}^{n} X_i!$$

Taking the log

$$\ell(\mu) := \log \mathrm{lik}(\mu) = \sum_{i=1}^{n} X_i \log(\mu) - n\mu - \sum_{i=1}^{n} \log(X_i!)$$

and taking a derivative

$$\ell'(\mu) = \frac{1}{\mu} \sum_{i=1}^{n} X_i - n$$

and solving this equation yields

$$\widehat{\mu}_{MLE} = \overline{X}_n$$

Formlally, one needs to verify that this solution is a maximum, not a minimum. I leave that to you. We observe that in this case, the MLE coincides with the MOM.

Normal $N(\mu, \sigma^2)$

Here the likelihood function is a function in two arguments

$$\begin{aligned} \mathrm{lik}(\mu, \sigma) &= \prod_{i=1}^{n} \left\{ \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2} \left(\frac{X_i - \mu}{\sigma} \right)^2 \right) \right\} \\ &= (2\pi)^{-n/2} \sigma^{-n} \exp\left(-\frac{1}{2} \sum_{i=1}^{n} \left(\frac{X_i - \mu}{\sigma} \right)^2 \right) \end{aligned}$$

and we get

$$\ell(\mu, \sigma) := \log \mathrm{lik}(\mu, \sigma) = -\frac{n}{2} \log(2\pi) - n \log(\sigma) - \frac{1}{2} \sum_{i=1}^{n} \left(\frac{X_i - \mu}{\sigma} \right)^2$$

Taking partial derivatives gives the MLEs:

$$\frac{\partial \ell(\mu, \sigma)}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^{n} (X_i - \mu)$$

and

$$\frac{\partial \ell(\mu, \sigma)}{\partial \sigma} = -\frac{n}{\sigma} + \sigma^{-3} \sum_{i=1}^{n} (X_i - \mu)^2$$

Solving

$$\begin{aligned} \left.\frac{\partial \ell(\mu,\sigma)}{\partial \mu}\right|_{\mu=\widehat{\mu}_{MLE},\ \sigma=\widehat{\sigma}_{MLE}} &= 0 \\ \left.\frac{\partial \ell(\mu,\sigma)}{\partial \sigma}\right|_{\mu=\widehat{\mu}_{MLE},\ \sigma=\widehat{\sigma}_{MLE}} &= 0 \end{aligned}$$

yields

$$\widehat{\mu}_{MLE} = \overline{X}_n \qquad \text{and} \qquad \widehat{\sigma}^2_{MLE} = \widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2.$$

Once again, the MOM and MLE coincide.

Gamma(α, λ)

Recall that the density is given by

$$f(x|\alpha,\lambda) = \frac{1}{\Gamma(\alpha)}\lambda^\alpha x^{\alpha-1}\exp(-\lambda x),\ x>0$$

Hence,

$$\begin{aligned} \text{lik}(\alpha,\lambda) &= \Gamma^{-n}(\alpha)\lambda^{n\alpha}\prod_{i=1}^{n} X_i^{\alpha-1}\exp(-\lambda\sum_{i=1}^{n}X_i) \\ \ell(\alpha,\lambda) &= \alpha n\log(\lambda) + (\alpha-1)\sum_{i=1}^{n}\log(X_i) - \lambda\sum_{i=1}^{n}X_i - n\log\Gamma(\alpha) \end{aligned}$$

Taking the derivative with respect to λ yields

$$\frac{\partial \ell(\alpha,\lambda)}{\partial \lambda} = \frac{\alpha n}{\lambda} - \sum_{i=1}^{n} X_i$$

so that

$$\widehat{\lambda}_{MLE} = \frac{\widehat{\alpha}}{\overline{X}_n}$$

However, we do not have a closed form solution for α as

$$\frac{\partial \ell(\alpha,\lambda)}{\partial \alpha} = -n\frac{\Gamma'(\alpha)}{\Gamma(\alpha)} + n\log\lambda + \sum_{i=1}^{n}\log X_i$$

Setting this equation to zero for $\alpha = \widehat{\alpha}_{MLE}$ and $\lambda = \widehat{\lambda}_{MLE} = \widehat{\alpha}_{MLE}/\overline{X}_n$ means that we need to solve the equation

$$-n\frac{\Gamma'(\alpha)}{\Gamma(\alpha)} + n\log\alpha - n\log\overline{X}_n + \sum_{i=1}^{n}\log X_i = 0.$$

We need to rely on numerical solutions to find the implicit solution to this equation. As starting values, we could take the (consistent) MOMs.

Uniform$(0,\theta)$

As the density is $f(x|\theta) = 1/\theta$, the likelihood function equals θ^{-n}. Maximizing this function gives $\theta = 0$, right? Wrong!

This is an example where the domain of the density depends on the parameter. This requires extra care. The density can be written as

$$f(x|\theta) = \theta^{-1}1\{0 \le x \le \theta\}$$

so, taking the indicator into account, we have

$$\begin{aligned}\text{lik}(\theta) &= \theta^{-n}1\{X_1 \le \theta\}1\{X_2 \le \theta\}\cdots 1\{X_n \le \theta\}\\ &= \theta^{-n}1\{X_{(n)} \le \theta\}\\ &= \theta^{-n}1\{\theta \ge X_{(n)}\}\end{aligned}$$

Here $X_{(n)} := \max(X_1,\ldots,X_n)$. So draw a graph of this likelihood as a function in θ. You see that it is zero till $\theta = X_{(n)}$, then it decreases rapidly as θ^{-n}. Clearly, this function is maximized at $\theta = X_{(n)}$, that is, we argued that

$$\widehat{\theta}_{MLE} = X_{(n)}$$

This makes sense: all observations X_i must in the interval $(0,\theta)$. The largest one is closest to θ and the most likely value. All previous examples gave estimates that were

smooth functions of $\overline{X}_n$. This case results into $X_{(n)}$, and maxima and means show different behavior. We will see in an exercise that the rate of convergence of the MSE is faster than before - $O(1/n^2)$ versus $O(1/n)$, and its limiting distribution is quite different too. More precisely, you will show that

$$\mathbb{E}[\widehat{\theta}_{MLE}] = \frac{n}{n+1}\theta$$

which is strictly less than θ, for each fixed sample size n. Again, we only get observations that are no larger than θ, so we are always biased below. Asymptotically, as $n \to \infty$, $\mathbb{E}[\widehat{\theta}_{MLE}] \to \theta$. In addition, the MSE equals $2\theta^2/\{(n+1)(n+2)\}$, so consistency follows as well. Compared to the MOM, the MLE has a superior MSE, it differs by a factor $1/n$.